

Statistical Foundation for Data Science

Time: 3 Hours

Max Marks: 75

Section – A

(Short Answer Questions – Answer any FIVE questions. Each carries 5 marks)

5 × 5 = 25 Marks

1. Define Statistics. Mention its scope in data science.
2. Explain the difference between primary and secondary data.
3. Describe the merits and demerits of mean, median, and mode.
4. What is standard deviation? How is it useful?
5. Define correlation. How is it measured?
6. What is regression? Differentiate between correlation and regression.
7. State the addition and multiplication theorems of probability.
8. What is a probability distribution? Explain with an example.

Section – B

(Essay Type Questions – Answer ALL questions. Each carries 10 marks)

5 × 10 = 50 Marks

9. (a) Define Statistics and explain its importance in data analysis and data science.
OR
(b) Discuss the types of data and methods of data collection in statistics.
10. (a) Calculate the mean, median, and mode for the following data:
10, 15, 20, 25, 25, 30, 35, 35, 35, 40
OR
(b) Explain the concept of dispersion and describe different methods to measure it.
11. (a) Explain Karl Pearson's coefficient of correlation with a suitable example.
OR
(b) Discuss the concept of linear regression. Explain how to predict a value using a regression equation.
12. (a) Define probability. Explain the classical and empirical definitions with examples.
OR
(b) Explain the concepts of conditional probability and Bayes' theorem with suitable examples.
13. (a) Describe Binomial and Poisson distributions. Mention their applications in data science.
OR
(b) Explain the Normal Distribution and its properties with a diagram.